PW SKILLS

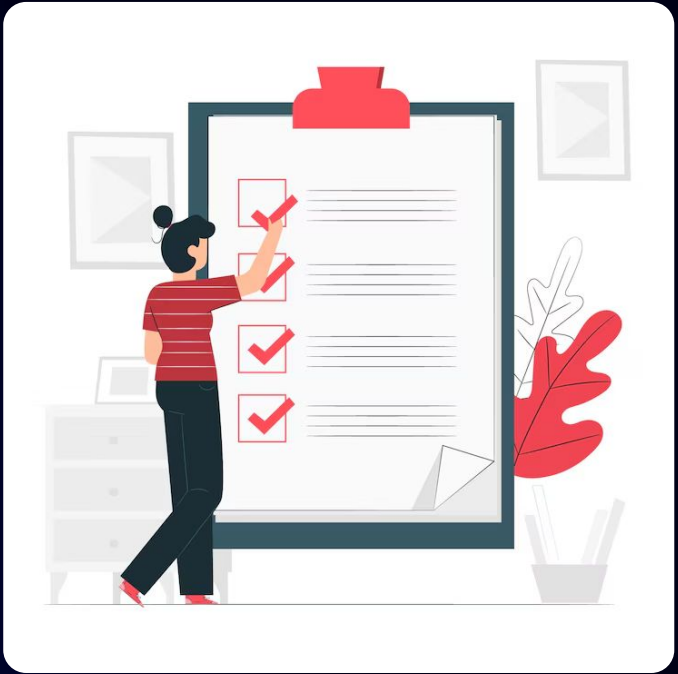
Loops & Problem Solving in JavaScript

Learning Objectives

- Use loops (for, while, do...while) for repetition
- Apply break and continue for control flow
- Understand and use switch statements
- Choose between switch and if-else
- Solve problems using loops and control statements

Prerequisites

- Basic understanding of programming concepts
- Familiarity with variables, data types, and conditions
- Basic knowledge of loops (for, while)
- Understanding of simple decision-making (if-else)
- Familiarity with writing and running code in an editor/IDE
- Basic debugging and problem-solving skills
- Willingness to practice and experiment with logic building



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Why Loops are Needed

Overview of Loops

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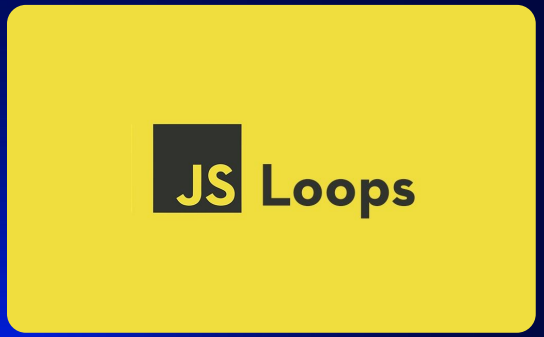
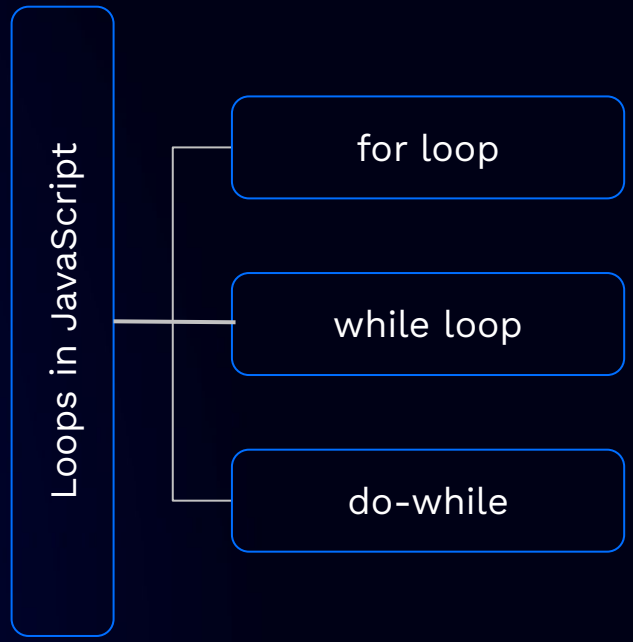
- Loops, or iteration statements, are essential in programming and critical to modern computing.
- They enable automation by allowing code to run repeatedly based on specific instructions.
- A loop is a block of code with instructions on how long to keep it running repeatedly.

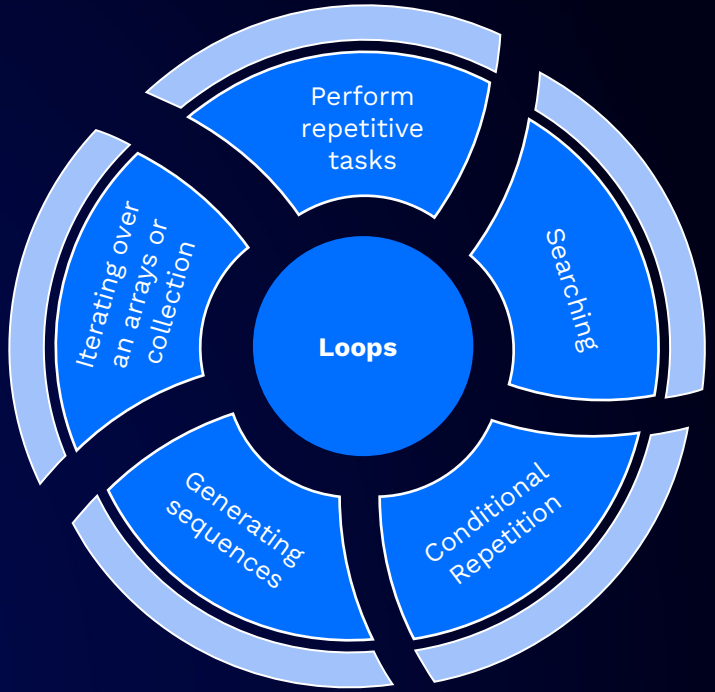


Types of loops in JavaScript

- The basic types of loops in JavaScript are
  - for loop
  - while loop
  - do-while loop



Usage of loops



Pop Quiz

Q. Select the basic types of loop in javascript

A

do while

B

If else

Pop Quiz

Q. Select the basic types of loop in javascript

A

do while

B

If else

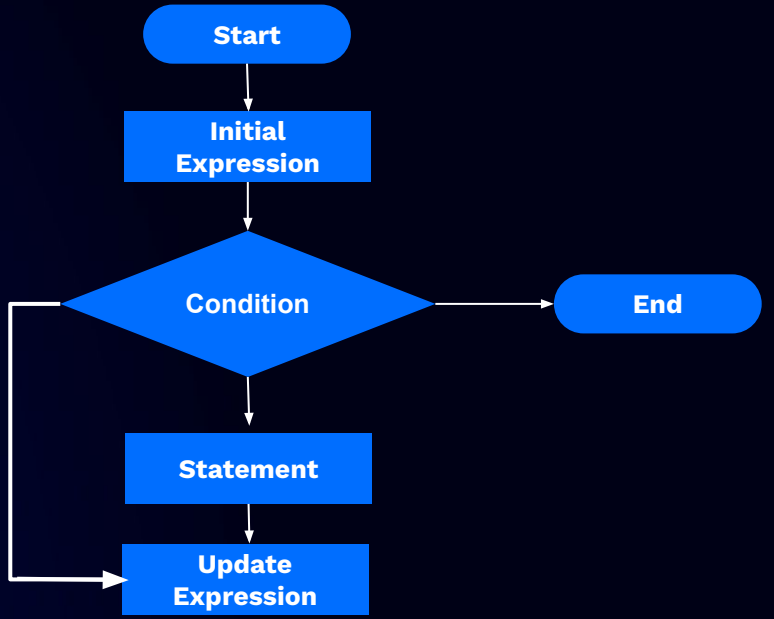
for loop

For loop in JavaScript

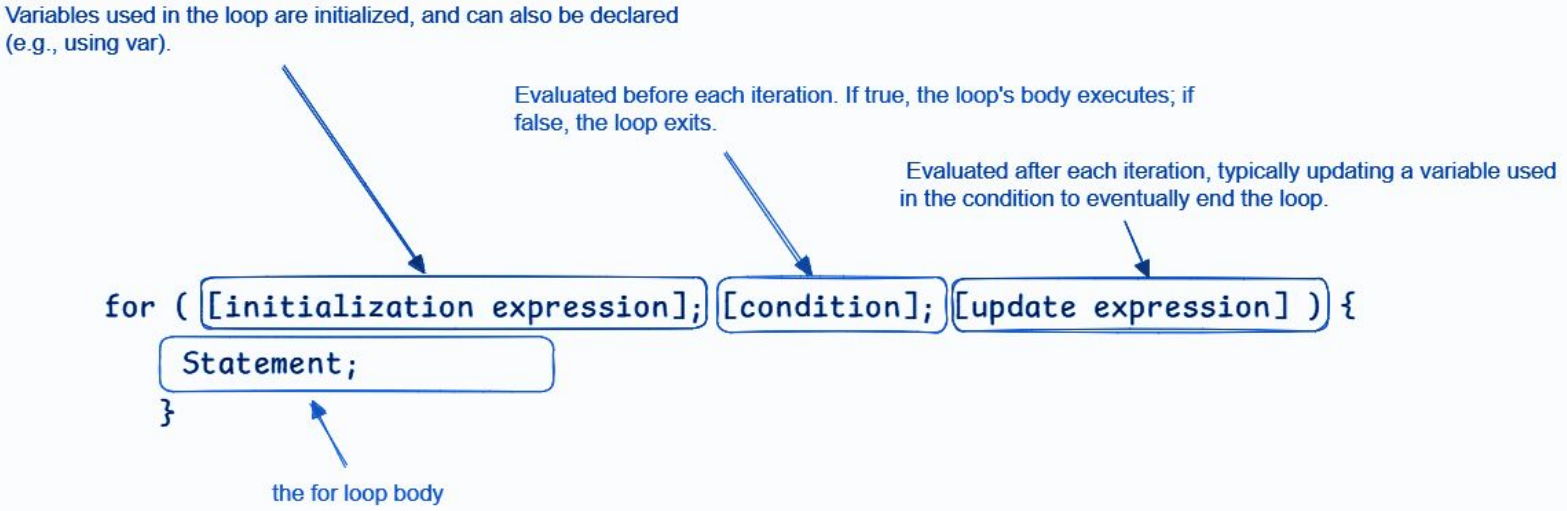
- The **for** loop is meant to repeatedly execute a piece of code a known number of times.
- We use **for** loop to run a piece of code until the set condition turns false
- Syntax -

```
for ([initialization expression]; [condition]; [update expression]){
      Statement;
}
```

Flow of the for loop



Understanding the for loop syntax



Hands-On Demo | for loop in JavaScript

Pop Quiz

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Q. Which statement is true about the for loop's update expression?

A

It is executed after the loop's body

B

It is executed before the loop's body

Pop Quiz

PW SKILLS

Q. Which statement is true about the for loop's update expression?

A

It is executed after the loop's body

B

It is executed before the loop's body

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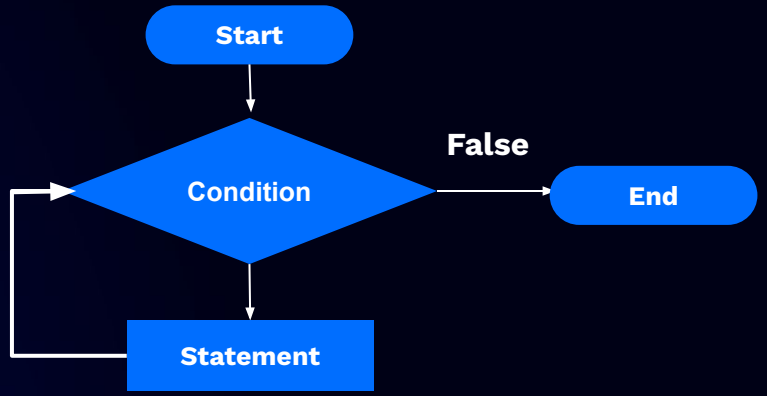
while loop

while loop in JavaScript

- The while loop is meant to repeatedly execute a piece of code an unknown number of times.
- This loop keeps running as long as the condition is true

Syntax -

```
while (condition) {
    // body of the loop
}
```



Flow of the while loop

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Hands-On Demo | while loop in JavaScript

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Q. What will be the output of the following code ?

```
let i = 0;
while (i < 3) {
    console.log(i);
    i++;
}
```

A

0 1 2 3

B

0 1 2

Pop Quiz

PW SKILLS

Q. What will be the output of the following code ?

```
let i = 0;
while (i < 3) {
    console.log(i);
    i++;
}
```

A

0 1 2 3

B

0 1 2

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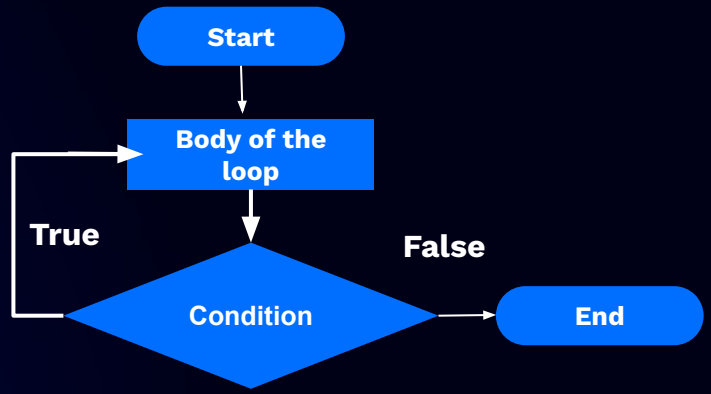
do-while loop

do-while loop in JavaScript

- Do while loop is similar to while loop with one difference that the 1st iteration runs always, and later iterations run after the condition evaluates to true
- Syntax -

Syntax -

```
do {
      // body of the loop
} while (condition)
```



Flow of the do-while loop

Comparison on for, while and do while loops

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Feature	for Loop	while Loop	do-while Loop
Syntax	for (initialization; condition; update) { // code }	while (condition) { // code }	do { // code } while (condition);
condition	Before each iteration	Before each iteration	After each iteration
Execution	Not guaranteed (may not execute if condition is false initially)	Not guaranteed (may not execute if condition is false initially)	Guaranteed to execute at least once

Comparison on for, while and do while loops

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Feature	for Loop	while Loop	do...while Loop
Use cases	When the number of iterations is known or can be easily determined	When the number of iterations is not known, and loop should execute as long as the condition is true	When the loop must execute at least once regardless of the condition
condition	Before each iteration	Before each iteration	After each iteration
Example	for (let i = 0; i < 5; i++) { console.log(i); }	let i = 0; while (i < 5) { console.log(i); i++; }	let i = 0; do { console.log(i); i++; } while (i < 5);

Hands-On Demo | do-while loop

Pop Quiz

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Q. What will be the output of the following code ?

```
let count = 0;
do {
    count++;
} while (count < 0);
console.log(count);
```

A

1

B

0

Pop Quiz

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Q. What will be the output of the following code ?

```
let count = 0;
do {
    count++;
} while (count < 0);
console.log(count);
```

A

1

B

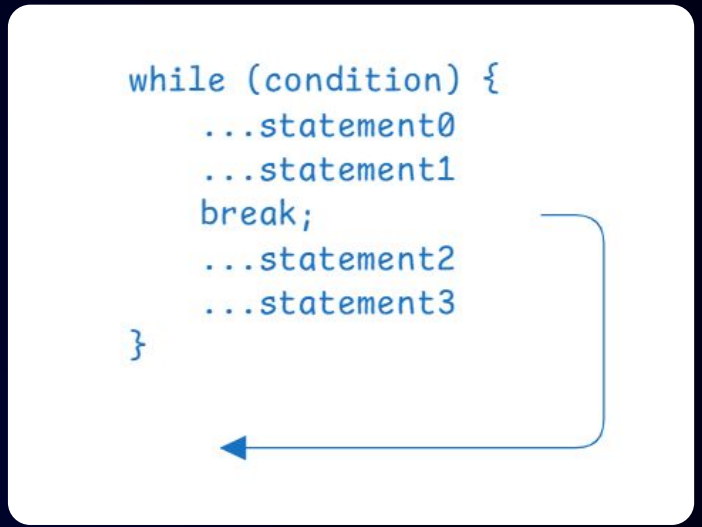
0

Break and Continue

Break statement in JavaScript

- break statement, without a label reference, can only be used to jump out of a loop or a switch block.
- It terminates the current loop or switch statement and transfers program control to the statement following the terminated statement.
- **Syntax -**

```
break;
```

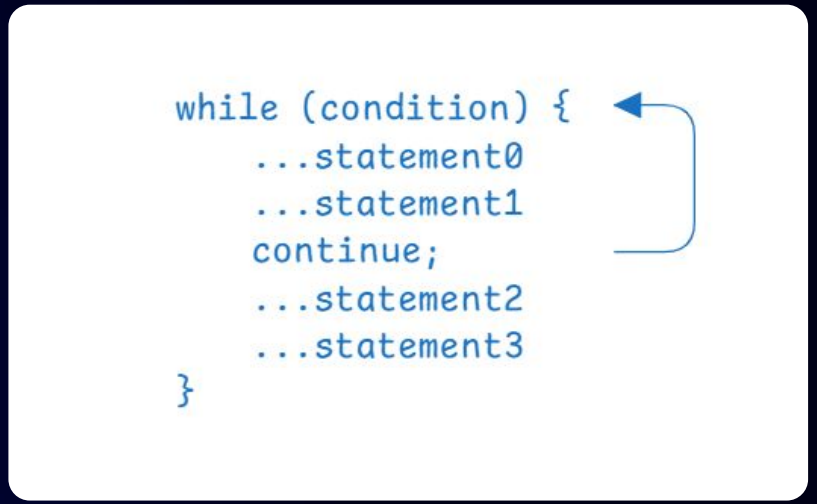


Continue statement in JavaScript

- **continue** statement, with or without a label reference, can only be used to skip one loop iteration.

- Skips the execution of the statements in the current iteration of the current or labeled loop, and continues execution of the loop with the next iteration.
- **Syntax -**

  continue;



Comparing break and continue statements

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Statement	Purpose	Usage Example	Typical use case
continue	Skips the rest of the code inside the loop for the current iteration and moves to the next iteration	javascript for (let i = 0; i < 5; i++) { if (i === 2) continue; console.log(i); } // Output: 0 1 3 4	When you need to skip certain iterations in a loop
break	Exits the loop immediately, and the control moves to the code following the loop	javascript for (let i = 0; i < 5; i++) { if (i === 2) break; console.log(i); } // Output: 0 1	When you need to terminate a loop based on a condition

Hands-On Demo | Break and Continue Statements

Pop Quiz

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Q. What will be the output of the following code ?

```
for (let i = 0; i < 5; i++) {
    if (i === 3) break;
    if (i === 1) continue;
    console.log(i);
}
```

A

0 1 2

B

0 2

Pop Quiz

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Q. What will be the output of the following code ?

```
for (let i = 0; i < 5; i++) {
    if (i === 3) break;
    if (i === 1) continue;
    console.log(i);
}
```

A

0 1 2

B

0 2

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Take A 5-Minute Break!

- Stretch and relax
- Hydrate
- Clear your mind
- Be back in 5 minutes

Switch Deep Dive

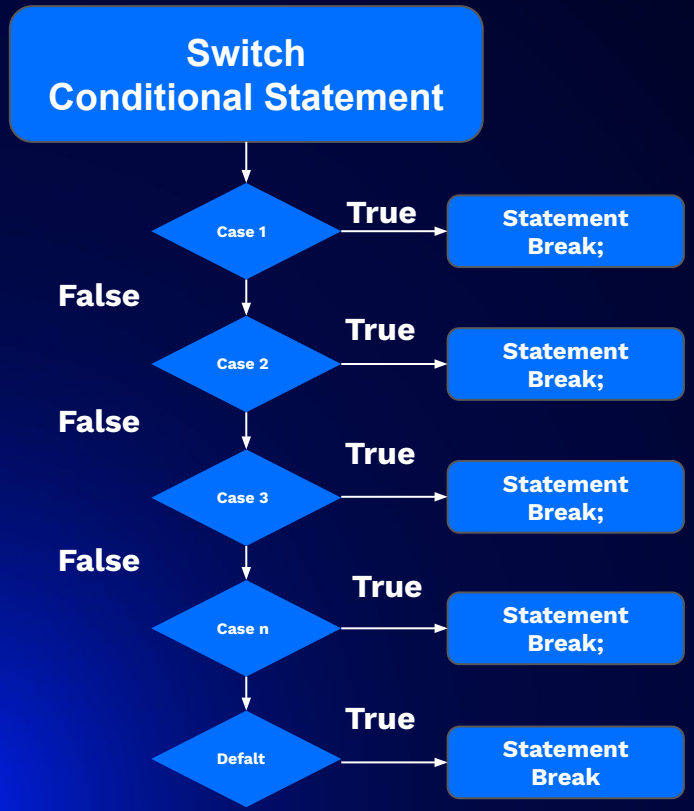
Introduction to Switch statement and Syntax

- The switch statement evaluates an expression.
- It matches the expression's value against case clauses.
- Executes statements after the first matching case until a break statement is encountered.
- The default clause executes if no case matches the expression's value.

```
switch (expression) {
  case value1:
    //Statements executed when the result of expression matches value1
    break;
  case value2:
    //Statements executed when the result of expression matches value2
    break;
  ...
  case valueN:
    //Statements executed when the result of expression matches valueN
    break;
  default:
    //Statements executed when none of the values match the value of the expression
    break;
}
```

Syntax of the switch statement

Flow of the Switch statement



Some of the usage cases of Switch statement

- When you need to perform different actions based on user input or choices.
- Executing different code based on the day of the week.
- Managing application states or modes.
- Managing access to different parts of an application based on user roles.
- Handling different game states or player actions.

Comparison of the Switch and if statements

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Feature	Switch Statement	if Statement
Purpose	Best for comparing a single expression against multiple possible values.	Best for evaluating multiple different or complex conditions.
Syntax Readability	More readable for simple value comparisons.	More flexible for complex logical conditions.
Fall-Through Behavior	Executes all cases after a matching one unless break is used.	Executes only the matched block; no fall-through.

Comparison of the Switch and if statements

Feature	Switch Statement	if Statement
Performance	Generally faster for multiple discrete value comparisons.	Potentially slower with many conditions due to multiple evaluations.
Default Handling	Uses default clause for unmatched cases.	Uses else clause for unmatched conditions.

Hands-On Demo | Switch Statements

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Q. What will be the output of the following code ?

```
let text = 'JavaScript';
switch (text[1]) {
   case 'a':
   case 'e':
   case 'i':
   case 'o':
   case 'u':
      console.log('Vowel');
   default:
      console.log('Consonant');
}
```

A

Vowel
Consonant

B

vowel

Pop Quiz

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Q. What will be the output of the following code ?

```
let text = 'JavaScript';
switch (text[1]) {
   case 'a':
   case 'e':
   case 'i':
   case 'o':
   case 'u':
      console.log('Vowel');
   default:
      console.log('Consonant');
}
```

A

Vowel
Consonant

B

vowel

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Logical Problem Solving Patterns

Collaboration Workflow Using Branches

In team projects, multiple developers work together 👥
Best practice:

- ❌ Do NOT work directly on `main`
- ✅ Use separate branches for features

Helps:

- Avoid conflicts
- Keep main branch stable

Commonly used on GitHub

Pop Quiz

Q. What is a limitation of using GUI tools like GitHub Desktop?

A

Deep understanding of Git concept

B

Less understanding of Git concepts

Practical Case Studies (Hands-On)

- print even numbers from 1 to n
- sum of numbers from 1 to n
- count numbers divisible by 3
- simple menu using switch

Summary

In this lecture, we have covered:

- Understand the need for loops in programming
- Use for, while, and do...while loops effectively
- Apply break and continue for control flow
- Use switch for decision-making
- Solve basic logical problems using loops
- Implement practical programs (number printing, sum, divisibility, menu systems)

DOUBT RESOLUTION

Important

- Complete the post-class assessment
- Complete assignments (if any)
- Practice the concepts and techniques taught in this session
- Review your lecture notes
- Note down questions and queries regarding this session and consult the teaching assistants.

THANK YOU